

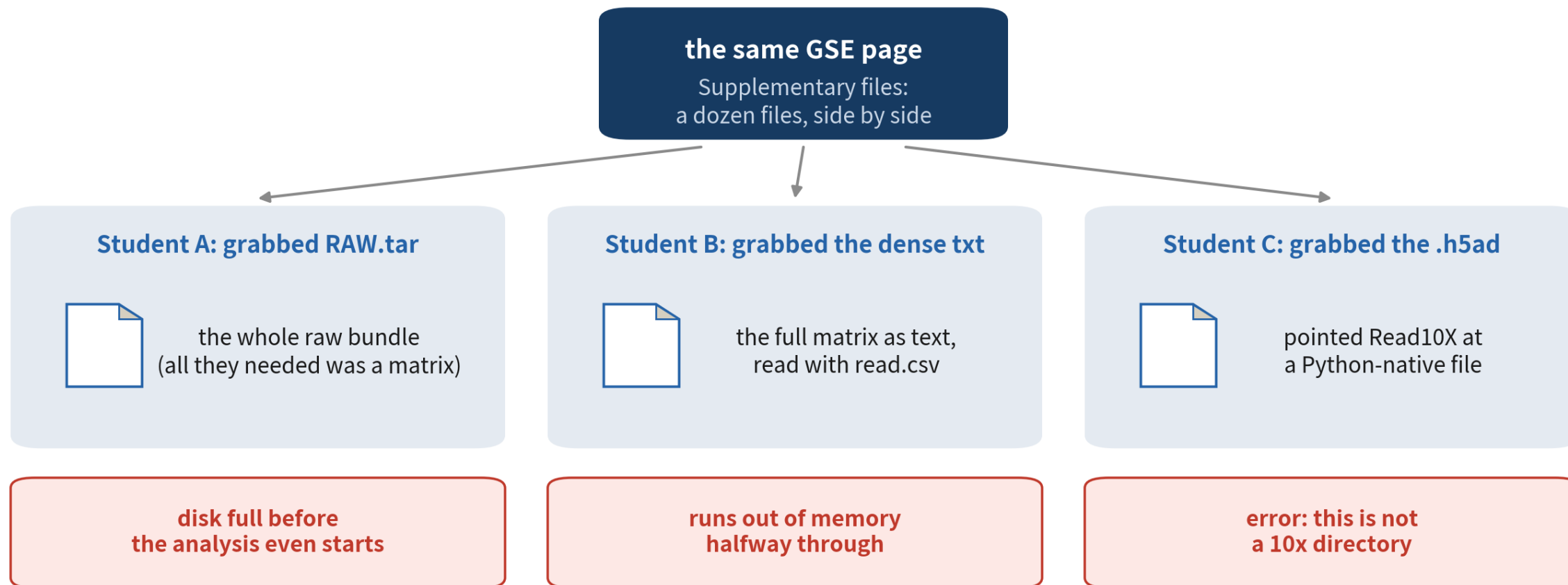
BS1 Series B

Getting public data, dissecting formats: the real-world workflow

From an accession code to an analysable matrix — the data map, download discipline, seven formats, one scale-reading routine.

About 35 min | Prerequisites: B1, B5 | Series B Supplement, episode 1

One GSE page, three people, three files



schematic

nothing is broken and nobody lied — all three simply failed to recognise the file

All three are stuck — yet nothing is broken and nobody lied

WHERE THIS EPISODE STARTS

**It's not the data that stops you —
it's not recognising what it looks like.**

The skill isn't 'knowing how to download' — it's recognising every file and knowing which key opens it. This episode is that keyring.

What you will learn

- 1 Read the public-data map: three entry layers, the accession table, and the raw vs processed split
- 2 Build download discipline: GEO navigation, wget -c, md5 checks, and detouring around controlled access
- 3 Dissect seven matrix formats, read three value scales, and log every decision in decisions.md

01

The public-data map

Know the landscape before you download anything

Three layers of entry points

Three layers of public-data entry points

higher = authoritative and complete; lower = convenient — search bottom-up

Layer 1 — repositories (the source of record)

GEO

>200k studies (NAR 2024)

BioStudies

home of E-MTAB (European)

GSA

CRA open / HRA controlled

SRA

raw FASTQ territory

Layer 2 — atlas & curation portals (browse first)

CELLxGENE

99.6M unique human cells

EBI SCEA

one pipeline, 10M cells

UCSC CB / SCP

authors' own UMAPs + labels

HCA / DISCO

atlas projects / deep integration

Layer 3 — one-line loaders (class & practice)

SeuratData

pbmc3k, ifnb, ...

scRNAseq (Bioc)

fetchDataset() returns SCE

sc.datasets

Python built-in teaching sets

10x Datasets

official demos (CC BY 4.0)

search lower first ↓

Repositories are authoritative, portals let you browse first, loaders are effortless — search bottom-up

How to use the three layers

Numbers from the reference library's 2026-09 verification snapshot; confirm on the official pages

Layer	Representatives	When to use it
Repositories	GEO (>200k studies), BioStudies, GSA	You hold an accession; home of nearly every problem-set item
Atlas / curation portals	CELLxGENE (99.6M unique cells), EBI SCEA, UCSC CB, Broad SCP	Finding new data; inspect authors' labels and UMAPs before committing
One-line loaders	SeuratData, scRNAseq, sc.datasets, 10x Datasets	Class demos and practice: old friends like pbmc3k and ifnb

Read the accession, know the door

Read the accession, know the door

when a paper gives you only a code, the prefix tells you where to go

GSE12345	→	GEO	processed matrices in Supplementary
SRR / SRP	→	SRA	raw FASTQ — reanalysis doesn't need it
E-MTAB-1234	→	BioStudies / ArrayExpress	European labs' default; direct FTP
SCP259	→	Broad Single Cell Portal	download may require Google sign-in
CRA001160	→	GSA (CNCB)	open, but mostly FASTQ
HRA...	→	GSA-Human	controlled access — detour on sight
E-CURD-46	→	EBI SCEA	matrices requantified on one pipeline



When a paper hands you only a code, the prefix is the signpost — keep this table

The accession lookup table

One study may carry several codes — pick the processed one

Prefix	Where it lives	Watch out
GSE	GEO	processed files sit at the bottom, under Supplementary
SRR / SRP	SRA	raw FASTQ; reanalysis doesn't need it
E-MTAB	BioStudies / ArrayExpress	the European default; direct FTP, no login
SCP	Broad Single Cell Portal	download may require a Google sign-in
CRA	GSA (CNCB)	open, but mostly FASTQ
HRA	GSA-Human	controlled access — detour on sight
E-CURD	EBI SCEA	requantified, cross-comparable matrices

Raw vs processed: the two-layer rule

One study, two layers of data

reanalysis needs only the processed layer — never apply for controlled access just to get FASTQ

raw layer: FASTQ reads

- lives in SRA / EGA / dbGaP / GSA
- tens to hundreds of GB; often controlled
- needed only to re-align (B2–B3)



**see dbGaP / HRA /
DUOS / EGA?**

that's the raw-access gate.
Detour: GEO's processed files,
or the Data availability section
→ Zenodo / Dryad / figshare

processed layer: count matrices

- lives in GEO Supplementary, portals, journal files
- MB to GB scale; mostly login-free
- the problem sets and most reanalyses need only this



record three things per download

portal + release version
accession
download date

Raw lives behind gates at tens of GB; processed lives in Supplementary, mostly login-free

THE RULE OF THIS SECTION

Reanalysis needs the processed layer. Only.

B2-B3 covered re-running from FASTQ — small demo data suffices there. Never apply for controlled access just to fetch FASTQ.

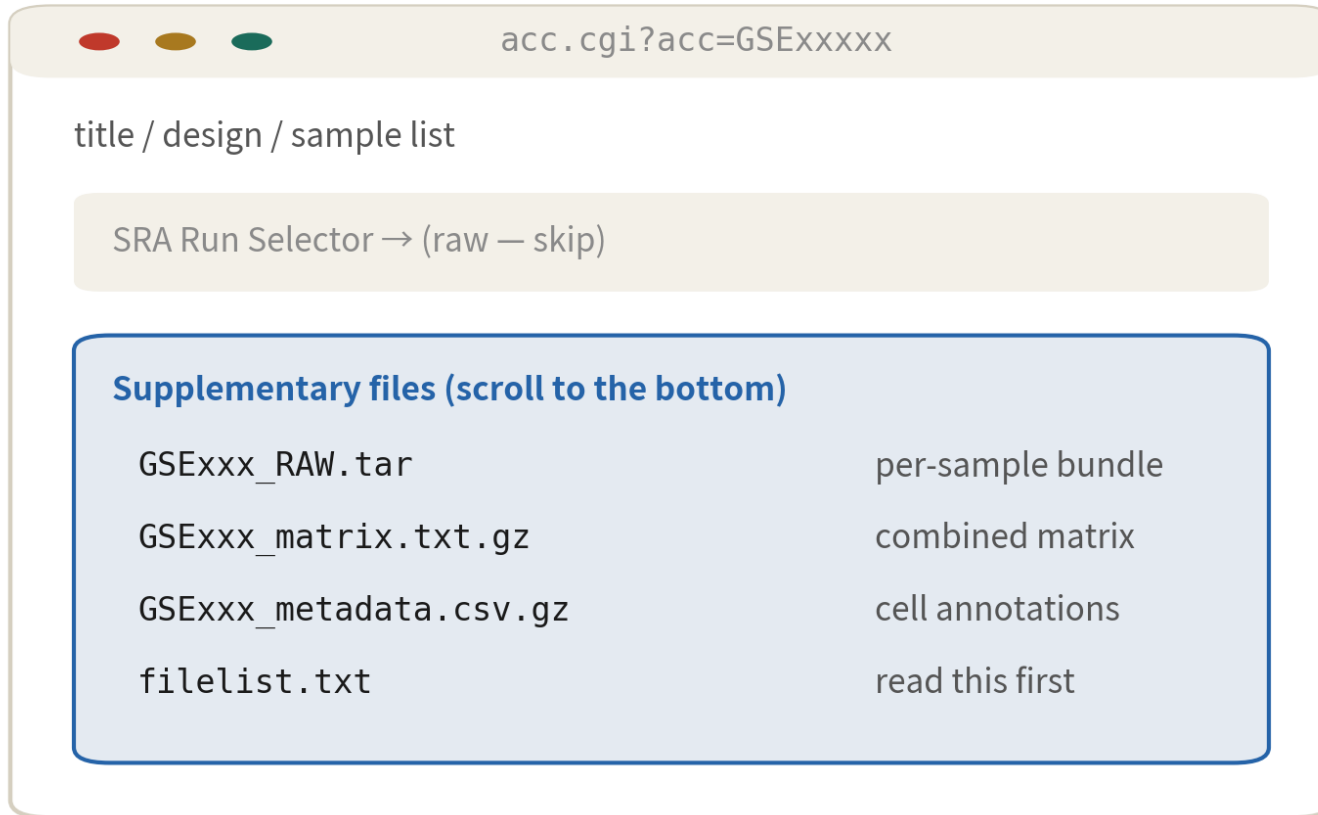
02

Download practice

GEO navigation, resumable transfers, md5, big-file strategy, spotting the gates

Navigating a GSE page

Navigating a GSE page; taming big files



acc.cgi?acc=GSExxxxx

[title / design / sample list](#)

SRA Run Selector → (raw — skip)

Supplementary files (scroll to the bottom)

GSExxx_RAW.tar	per-sample bundle
GSExxx_matrix.txt.gz	combined matrix
GSExxx_metadata.csv.gz	cell annotations
filelist.txt	read this first

1. smallest file first

metadata + filelist to confirm, then commit to the big one

2. wget -c to resume

broken transfers resume; reserve disk space first

3. md5sum to verify

a download isn't done until the md5 matches

Open `acc.cgi?acc=GSE...` and scroll to the very bottom — Supplementary files is your destination

The download trio: wget -c + md5sum

```
# 大檔斷點續傳：斷線重跑同一行就會接著抓
wget -c "https://ftp.ncbi.nlm.nih.gov/geo/.../GSExxx_RAW.tar"

md5sum GSExxx_RAW.tar # 跟頁面 /filelist 的 md5 對
tar -tf GSExxx_RAW.tar | head # 解開前先看內容清單
```

```
a3f9...e21c GSExxx_RAW.tar ← 對上才算下載完成
GSM000001_sample1.matrix.gz
GSM000001_sample1.features.tsv.gz
GSM000001_sample1.barcodes.tsv.gz
```

-c means continue: no restarts from zero; if the md5 mismatches, re-download — never force-read

Three moves for big files

Smallest file first



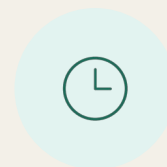
Confirm with metadata
+ filelist before
committing to the big
matrix (same discipline
as spatial data)

RAW.tar sells by the piece



Per-sample bundles
need not be carried
whole — grab one
sample's file from its
GSM page

Budget disk before fetching



The problem sets' rule
of thumb: 5–10 GB per
exercise; unpacking
and converting need
workspace too

Controlled access: just learn the door signs

What it is

dbGaP, EGA, DUOS, Synapse, HRA — when these names appear, that data needs applications, approvals, agreements.



Why it happens

They guard the privacy risk of raw sequencing data; the processed matrix usually has a login-free home elsewhere.

What it costs you

The right response is not to apply but to detour: GEO's processed files, or the paper's Data availability → Zenodo / Dryad / figshare.

The detour in practice: a real case

Front door: GSA has only FASTQ



PDAC CRA001160 — raw files need a login and are all FASTQ (the scenario of advanced exercise 13)

Read Data availability



The authors parked a processed h5ad on Zenodo (record 3969339) — direct, login-free

Verify the detour too



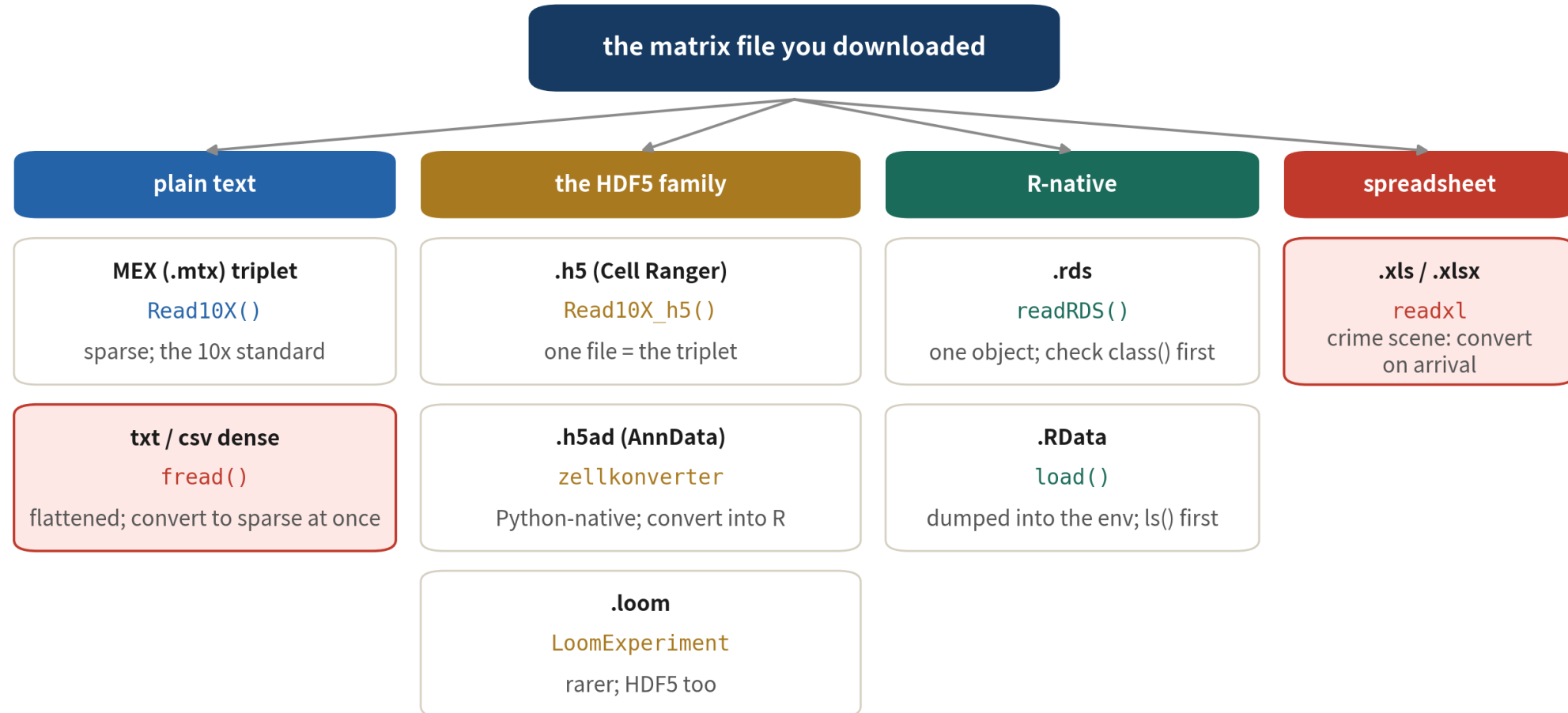
That file is mislabelled CRC but contains PDAC — run this episode's five questions on it anyway

03

Format dissection

The core of this episode: seven outfits, one way to recognise them

The format family tree



one biological payload, seven outfits — recognise the outfit to pick the key

The payload is always genes × cells of numbers — only the wrapping and the unwrapping function differ

Refresher: the 10x triplet (MEX)

matrix.mtx.gz



The sparse matrix
itself: coordinates and
values of non-zeros
only (B3's output)

features + barcodes



Row and column
names stored
separately — all three
files or nothing

Read10X eats the folder



From B5: pass the
folder path, not a file;
GEO copies often need
renaming first

HDF5 flavour one: Cell Ranger's .h5

```
library(Seurat)
# 單一 .h5 檔=三件套打包 (腎癌 GSE159115 就給這種)
m at<-Read10X_h5("GSM_tumor_filtered_feature_bc_matrix.h5")
class(m at)
obj<-CreateSeuratObject(m at,min.cells = 3)
```

```
[1] "dgCM atrix"
attr("package")
[1] "M atrix"
```

It loads straight into a sparse matrix — .h5 is the one-file triplet, not another world

The three HDF5 siblings

.h5 (Cell Ranger)



The triplet in one file;
Read10X_h5 handles it
in one line

.h5ad (AnnData)



The Python
ecosystem's standard
object — matrix,
metadata, embeddings
inside; needs
conversion into R

.loom



An older, rarer cousin,
HDF5 too;
LoomExperiment reads
it

h5ad into R: zellkonverter

```
library(zellkonverter)
# 基礎 20 同款: CELLxGENE 下載的 Tabula Sapiens 胰臟
sce <- readH5AD("TS_pancreas.h5ad") # 回 SingleCellExperiment
assayNames(sce) # 先看有哪些矩陣層
library(Seurat)
obj <- as.Seurat(sce, counts = "X", data = NULL)
```

```
[1] "X"
```

注意: X 是 counts 還是 log 過的值? ——讀檔五問的第 4 問

The conversion trap isn't the function — it's what scale lives in X; section four is devoted to it

h5ad ↔ Seurat: routes and reality

Main road: zellkonverter



Bioconductor, via the
SCE relay — the
course's and problem
sets' recommended
route

Alternative: SeuratDisk



Direct conversion via
h5Seurat; you babysit
the version
compatibility

The trend: rds is bowing out



CELLxGENE's rds
support is being
deprecated — R users
will need h5ad sooner
or later; practise now

Dense text matrices: fread, then sparse at once

```
library(data.table); library(Matrix)
x <- fread("GSExxx_counts.txt.gz") # dense: 比 read.csv 快
genes <- x[[1]]; x[,1] := NULL
m <- as.matrix(x); rownames(m) <- genes
sm <- as(m, "CsparseMatrix") # 立刻轉稀疏
rm(x, m); gc() # 放掉 dense, 高峰才會退
```

```
# 高峰出現在轉換瞬間: dense 與稀疏短暫並存
# 所以「讀得進來」的標準是: 記憶體 > dense 大小 + 稀疏大小
```

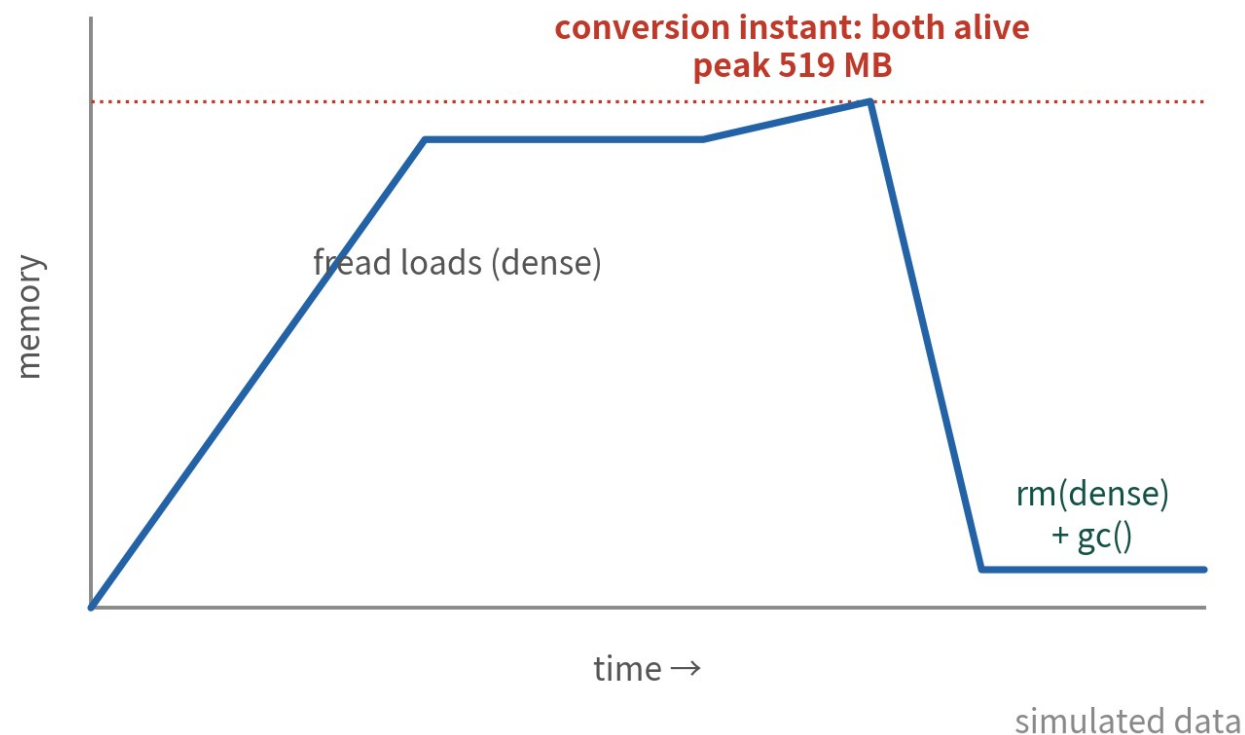
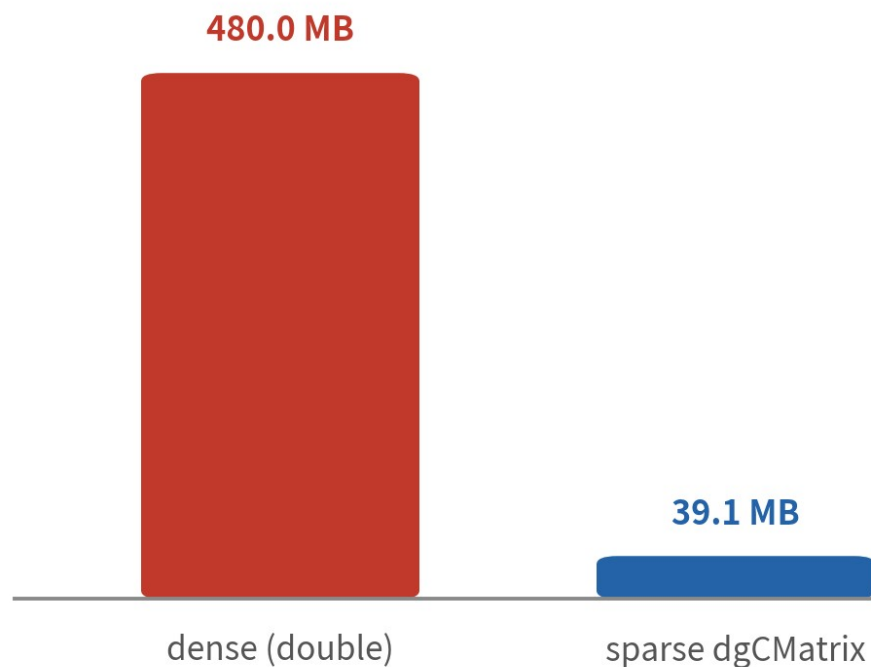
NPC GSE150430 and cSCC GSE144236 both ship as one big dense txt like this

Why convert at once: measured

Dense vs sparse storage, actually measured

simulated counts, 20,000 genes $\times$ 3,000 cells: 95% zeros

the same matrix, 12x smaller



The same simulated matrix: dense 480 MB, sparse 39 MB, conversion peak 519 MB — estimate the peak before you start

R-native files: rds and RData

rds: class() first



readRDS returns one object — a matrix, Seurat, SCE, or a Monocle3 cds (mouse MOCA ships as cds)

RData: ls() first



load() dumps objects into your environment — ls() before and after; the cell-line mixture RData drops three objects

Version caveat



Old objects often need UpdateSeuratObject; when a major version chokes, rebuild from the stored matrix layers

The XLS disaster

What it is

Whole count matrices saved as Excel files — neuroblastoma GSE137804 ships per-sample XLS.



Why it happens

Excel takes liberties: gene names like SEPT1 and MARCH1 famously become dates; reading is slow and memory-hungry.

What it costs you

The discipline: read with readxl, audit the gene names, convert to rds at once — the pipeline never touches XLS again.

Batch conversion with readxl

```
library(readxl)
fs <- list.files("GSE137804", "\\xls$", full.names = TRUE)
for (f in fs) {
  m <- read_excel(f)          # 一個樣本一個 XLS
  stopifnot(!any(grepl("^ [0-9]+ -", m[[1]]))) # 基因名變日期?
  saveRDS(m, sub("\\xls$", ".rds", f))      # 轉存，之後不碰 XLS
}
```

That stopifnot is a sentry: the moment a gene looks like '3-Sep', everything halts

The last trap: annotation rows inside the matrix

Trap format: annotation rows inside the matrix

the GSE72056 pattern — the first rows are not genes, they are per-cell annotations

Cell	cell_1	cell_2	cell_3	...
tumor	53	53	81	...
malignant (1/2/0)	2	1	2	...
non-malignant cell type	0	1	0	...
A1BG	0.0	2.3	0.0	...
A1CF	0.0	0.0	1.1	...
A2M	5.8	0.0	0.0	...



annotation rows → metadata

tumor / malignant / type
become a data.frame,
split off from the matrix

remaining rows → expression

values already $\log_2(\text{TPM}/10+1)$
— the scale must be read too
(next section)

rows 2–4: cell annotations; genes start at row 5

run QC without splitting? three annotation rows contaminate every statistic

Melanoma GSE72056: the first three rows are tumor / malignant / cell type — split into metadata before building

The scout: readLines(n = 10)

```
# 任何陌生文字檔，先看前幾行再決定怎麼讀  
readLines("GSE72056_melanoma_matrix.txt.gz", n = 4)
```

```
[1] "Cell\tcy53.1\tcy53.2\t..."  
[2] "tumor\t53\t53\t..."  
[3] "malignant(1=no,2=yes,0=unresolved)\t2\t1\t..."  
[4] "non-malignant cell type\t0\t1\t..."
```

Four lines of output expose the whole trap — the best value-for-keystrokes line in this episode

FORMATS, SUMMED UP

**Look at the file first.
Then choose the function.**

readLines for text, class for R objects, assayNames for SCE — never feed an unscouted file into the pipeline.

04

Reading the scale

Counts, TPM, log — the matrix won't introduce itself, but its histogram will

Three faces of the same data

The same data in three scales

simulated data

histogram the values first — the scale confesses on its own

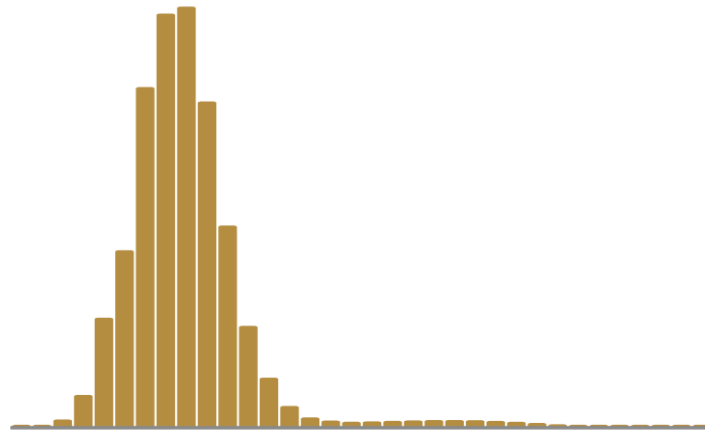
raw counts



value (log10 axis)

- all integers
- max 1,556
- 95% zeros

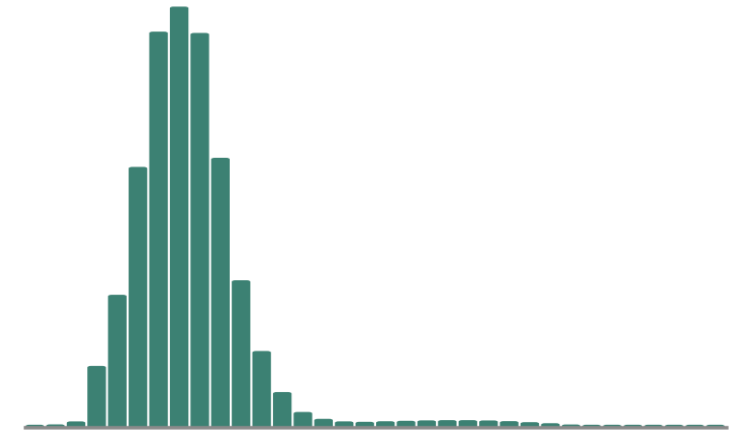
TPM / CPM (not logged)



value (log10 axis)

- decimals, no longer integers
 - max 220,031
- per-cell sum fixed at 1e6

$\log_2(\text{TPM}/10 + 1)$



value (linear axis)

- squeezed into single digits
 - max 14.4
- negatives? that's a different log

Integers with a long tail = counts; decimals with fixed column sums = TPM/CPM; squeezed into single digits = logged

The scale-reading cue table

The GEO page and Methods are testimony; the histogram is physical evidence — check both

Scale	Cues	Downstream action
raw counts	all integers, many zeros, max in the hundreds-plus	the normal route: start from NormalizeData
TPM / CPM	decimals; per-cell column sums nearly identical	library size already divided — don't divide again; take the log yourself
log scale	values squeezed into 0–low teens; $\log_2(\text{TPM}/10+1)$ is common	goes in the data slot, never log again; audit each method's scale assumption

Three lines to make the matrix confess

```
v <- sm @ x          # 稀疏矩陣的非零值
all(v == round(v))    # TRUE → 整數 → counts
max(v)                # 十幾 → 幾乎確定 log 過
head(Matrix::colSums(sm)) # 各欄總和幾乎相同 → TPM / CPM
```

```
[1] TRUE      ← 這份是 counts
[1] 1556
# 若 max 只有 14.4、又不是整數: log 尺度實錘
```

The 1556 and 14.4 in the output come from the simulated figure you just saw — three lines settle the case

The cost of misreading the scale

What it is

Stuffing a logged matrix into the counts slot, or running `NormalizeData` on TPM again.



Why it happens

Every downstream function assumes an input scale — treat that melanoma $\log_2(\text{TPM}/10+1)$ as counts and HVG plus DE all bend.

What it costs you

And nothing errors out. Wrong-scale plots still look lovely; the conclusions just can't be trusted — the same quiet failure as B19's disasters.

The file-reading decision log: decisions.md

What goes in



Every decision the data forced: format route, scale verdict, annotation-row handling, thresholds and reasons

A fixed format



What you found → the options → what you chose → why — same as the problem sets' real-data checkpoints

It is a deliverable



Advanced exercises grade it: this log is the draft of your Methods section

THE RULE OF THIS SECTION

**Misread the scale and everything downstream
bends —
silently.**

So the verdict needs physical evidence — the histogram, the three-line check — and the verdict itself goes into decisions.md.

05

Wrap-up: the five questions

The whole episode folds into one card

The five-question checklist

Five questions for every unfamiliar file

write every answer into decisions.md — it is the draft of your Methods section



1. Which layer is this?

raw or processed? reanalysis needs processed only



2. What format is it?

readLines(n=10) first, then choose the reader



3. Sparse or dense?

convert dense to sparse at once; estimate the peak first



4. What scale is it on?

counts / TPM / log? corroborate with a histogram



5. Any non-expression rows?

annotation rows, totals, duplicated genes — split before building

Run all five on every unfamiliar file, log the answers in decisions.md — pin this card next to your screen

The four sections, reviewed

Section	One line to keep
The data map	Read the accession to know the door; search the three layers bottom-up
Download practice	Small files first, wget -c, done only when md5 matches; detour at every gate
Format dissection	Recognise the outfit, pick the key; dense to sparse at once; XLS converted on arrival
Scale reading	The histogram confesses; verdicts and reasons all go into decisions.md

Where to go next

The reference library



The single-cell & spatial database reference: 16 first-line portals + 11 extensions, with a decision tree

The problem sets



20 basic + 60 advanced exercises; the advanced real-data checkpoints examine exactly this episode's skills

BS2: platform showdown



Next supplement: the world beyond 10x — how the platform reshapes QC and analysis logic

Check yourself 1

The paper's Data availability says only 'HRA004321'. Your next move?

- A File an access application on GSA-Human
- B Email the authors for the FASTQ
- C Recognise controlled access and detour to a processed copy (GEO / Zenodo / Dryad / figshare)
- D Give up on the dataset

Answer C. The HRA prefix means GSA-Human's controlled zone, and the gates guard raw data. Reanalysis needs processed only: search GEO, mine Data availability for general repositories — advanced exercise 13's Zenodo detour is the template.

Check yourself 2

fread just loaded a 2 GB dense txt matrix. The very next thing to do?

- A CreateSeuratObject right away
- B Convert to sparse, then rm the dense copy and gc()
- C Run NormalizeData first
- D Save a csv backup

Answer B. A flattened dense matrix can cost over ten times the memory (twelve-fold in our measurement), and both copies coexist at the conversion instant — convert, free the dense one, and everything downstream has room to run.

Check yourself 3

The matrix holds decimals, no negatives, max 14.4. The most likely scale?

- A raw counts
- B TPM (not logged)
- C a log scale (e.g. $\log_2(\text{TPM}/10+1)$)
- D z-score

Answer C. Counts are integers, unlogged TPM peaks in the tens of thousands, z-scores go negative — decimals squeezed into the low teens leave only the log family. Confirm, store in the data slot, never log again, and record the verdict in decisions.md.

NEXT EPISODE

BS2: Platform showdown — the world beyond 10x

Same blood, two platforms, wildly different QC plots — which one broke? Neither.

Droplets, microwells, full-length plates, combinatorial barcoding: the platform is part of your hypothesis. See you there.